

A Timestep Crossover Law for Stochastically Rounded Velocity Verlet on a Harmonic Oscillator

Cohen Brunner Thomas
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Abstract

We measure velocity-Verlet error for the unit harmonic oscillator, separating ordinary FP32 round-to-nearest from stochastic rounding. In the direct FP32 run, RMS final-state error has a grid minimum of 1.13×10^{-5} at $h = 1.53 \times 10^{-3}$ and rises by factors 2.5 and 4.4 at the next two finer steps. For stochastic rounding, we use the exact bias and variance decomposition and a common absolute timestep grid across durations. The grid-minimum precision exponent is $\hat{\beta}_s = 0.419$, with trajectory-bootstrap 95% interval $[0.392, 0.448]$, which contains the predicted $2/5$. An asymptotic fixed- B prediction, calibrated from $s = 10, 12, 20, 24$ without $s = 16$, predicts the observed grid minima at five $s = 16$ duration conditions to within 6.1%. The duration exponent is $\hat{\beta}_T = -0.205$, with 95% interval $[-0.254, -0.144]$, which contains $-1/5$. These results support the predicted scaling for this oscillator and implementation.

1 Introduction

For a fixed integration interval, reducing the step size decreases Verlet’s phase error but increases the number of floating-point updates. The final-state error therefore reflects both the deterministic error of the integrator and the accumulated effect of rounding. Its dependence on step size also depends on the rounding rule.

Floating-point effects in symplectic integration have been studied by Sanz-Serna and Calvo [1] and are treated in the broader framework of geometric numerical integration by Hairer, Lubich, and Wanner [2]. For stochastic rounding (SR), Connolly, Higham, and Mary prove mean-independence results for rounding errors and use them to derive probabilistic backward-error bounds for linear algebra [3]. Kehlet and Logg develop adjoint-weighted a posteriori ODE error estimates that include roundoff. Under their error model, the accumulated roundoff term scales inversely with the square root of the step size [4]. Croci et al. survey SR analysis and applications, including ODE computation [5].

Several studies examine probabilistic roundoff in ODE computation. Mosbach and Turner model accumulated rounding errors in deterministic ODE solvers and give trajectory distributions as functions of time, step size, and precision [6]. Croci and Giles analyze round-to-nearest and SR for Runge-Kutta finite-difference solutions of the heat equation and derive bounds on rounding-error accumulation [7]. Hopkins et al. study SR in reduced-precision fixed-point solvers for neuron models, reporting improved errors across several ODE solvers [8]. Hairer, McLachlan, and Razakarivony show that implementation choices affect long-time roundoff growth in Hamiltonian integration [9]. Their study concerns conservation errors for implicit Runge-Kutta schemes, whereas our measured quantity is the final-state error of velocity Verlet.

This paper tests a specific optimal-step law for velocity Verlet with stochastic rounding on the unit harmonic oscillator. We calculate the deterministic final-state bias from the exact Verlet matrix and estimate the covariance trace from independent trajectories. Their sum gives the mean-square risk without an independence assumption between truncation and rounding errors. We derive an asymptotic law from the leading Verlet phase term and a fitted covariance-growth model, then measure the precision exponent and test a fixed-coefficient prediction across duration. We fit the coefficient using significand widths 10, 12, 20, and 24, then evaluate its predictions at width 16. The contribution is this exact-bias and trajectory-covariance analysis with an absolute-scale transfer test for one oscillator, integrator, rounding implementation, and error metric. Its conclusions apply to that tested configuration.

We also report a separate FP32 round-to-nearest experiment. It shows a crossover for this setup, while the stochastic-rounding experiment isolates unbiased random rounding in a configurable binary format.

2 System and methods

We use the unit-frequency oscillator

$$H(q, p) = \frac{1}{2}(q^2 + p^2), \quad \dot{q} = p, \quad \dot{p} = -q,$$

with the exact solution available analytically. We write the state as $x = (q, p)^\top$ and the exact state after time T as $x(T) = \Phi(T)x_0$, where $\Phi(T)$ is the oscillator’s rotation matrix. The integration duration is $T = 100$ except in the duration sweep. Kick-drift-kick velocity Verlet uses N equal steps of size $h = T/N$. The direct FP32/FP64 comparison uses 128 initial phases on the unit-radius circle (energy $H = 1/2$). The controlled stochastic-rounding sweep fixes $x_0 = (1, 0)^\top$ and uses 128 independent trajectories for the precision sweep and 512 at each tested timestep in the duration sweep.

2.1 Direct standard precision

In the direct sweep, state values are rounded to IEEE FP32 or FP64 after each Verlet stage. This tests accumulated state rounding in software and does not emulate every hardware arithmetic pathway. For both precisions the sampled step counts are $N = 2^k$, $k = 6, \dots, 18$.

2.2 Stochastic rounding

The controlled sweep uses binary significand widths $s \in \{8, 10, 12, 16, 20, 24\}$. We use s for the number of significand bits and reserve p for momentum. Exponents are unbounded, isolating significand quantization. The coefficients h and $h/2$ are retained in FP64 to isolate rounding of the evolving state. At $T = 100$, the precision sweep uses integer step counts obtained by rounding $2^{k/2}$, for $k = 12, 13, \dots, 34$, and sets $h = T/N$. If an exact value z lies between adjacent representable numbers $z_- \leq z \leq z_+$, the operator returns

$$\text{SR}_s(z) = \begin{cases} z_-, & \text{with probability } (z_+ - z)/(z_+ - z_-), \\ z_+, & \text{with probability } (z - z_-)/(z_+ - z_-). \end{cases}$$

Thus $\mathbb{E}[\text{SR}_s(z) \mid z] = z$. Exactly representable values remain unchanged. A seeded independent random choice is made at each kick/drift state update (three rounding sites per step). This is actual stochastic rounding of state updates, not additive Gaussian noise. Arithmetic within each stage is evaluated before the state value is rounded. We use unit-roundoff scale $u = 2^{-s}$ and report ensemble RMS error over independent rounding trajectories.

The duration experiment holds $s = 16$ and uses the same widened absolute grid $h = 1/k$ for every duration, with $k = 35, 40, \dots, 200$. Each duration is an integer, so $N = Tk$ is integral and the grid is identical across all five durations. The grid is fixed in advance rather than centered on a theory-predicted optimum. The 170 duration/timestep conditions use distinct, reproducibly seeded pseudorandom streams.

The implementation and raw results are `experiment.py`, `precision_sweep.csv`, `stochastic_sweep.csv`, and `duration_sweep.csv`. Model fits are saved in `crossover_fit.json` and `duration_scaling.json`. The fixed-coefficient duration predictions are also available in `fixed_b_duration_predictions.csv`.

3 Quantitative crossover derivation

For unit frequency, velocity Verlet is the linear map

$$x_{n+1} = M_h x_n, \quad M_h = \begin{pmatrix} 1 - h^2/2 & h \\ -h(1 - h^2/4) & 1 - h^2/2 \end{pmatrix}.$$

The exact flow matrix is $\Phi(T) = \begin{pmatrix} \cos T & \sin T \\ -\sin T & \cos T \end{pmatrix}$. For the controlled sweep’s initial state $x_0 = (1, 0)^\top$, we take binary64 as reference arithmetic and treat the stage operations as exact relative to the simulated reduced precision. In this model, conditional unbiasedness at each stochastic-rounding site gives $\mathbb{E}[X_N] = M_h^N x_0$. Its bias relative to the exact solution is $b_h = M_h^N x_0 - \Phi(T)x_0$.

Within this reference-arithmetic model, the squared state error obeys the bias and variance identity

$$\mathcal{J}(h) = \mathbb{E}\|X_N - \Phi(T)x_0\|_2^2 = \|b_h\|_2^2 + \text{tr}(\Sigma_h), \quad \Sigma_h = \text{Cov}(X_N).$$

We compute the bias from the exact Verlet matrix and estimate the covariance trace from replicate trajectories, so no independence assumption between the two error sources is needed. For errors $e_j = X_N^{(j)} - \Phi(T)x_0$, $j = 1, \dots, J$, the finite ensemble identity is

$$\frac{1}{J} \sum_{j=1}^J \|e_j\|_2^2 = \|\bar{e}\|_2^2 + \frac{1}{J} \sum_{j=1}^J \|e_j - \bar{e}\|_2^2, \quad \bar{e} = \frac{1}{J} \sum_{j=1}^J e_j.$$

Here J is the number of replicate trajectories. We estimate the covariance with

$$\widehat{\Sigma}_h = \frac{1}{J-1} \sum_{j=1}^J (e_j - \bar{e})(e_j - \bar{e})^\top, \quad \widehat{\mathcal{J}}(h) = \|b_h\|_2^2 + \text{tr}(\widehat{\Sigma}_h).$$

The coefficient B is fitted to the covariance term alone.

Write $\theta_h = 2 \arcsin(h/2)$ and $\rho_h = \sqrt{1 - h^2/4}$. The matrix power for $x_0 = (1, 0)^\top$ has the closed form

$$M_h^N x_0 = (\cos(N\theta_h), -\rho_h \sin(N\theta_h))^\top.$$

The modified angular frequency and amplitude factor satisfy

$$\frac{\theta_h}{h} = 1 + \frac{h^2}{24} + O(h^4), \quad \rho_h = 1 - \frac{h^2}{8} + O(h^4).$$

For fixed T , the leading bias therefore contains both phase and amplitude terms:

$$b_h = h^2 \left(-\frac{T}{24} \sin T \cos T + \frac{1}{8} \sin T \right) + O_T(h^4).$$

Its exact leading coefficient is the norm of the displayed vector. For large T , this coefficient is $T/24 + O(1)$. We use the phase-only coefficient $A_{\text{ph}}(T) = T/24$ for the simple asymptotic prediction reported below. The full matrix bias enters every estimated risk and grid minimum. Across the five tested durations, the exact leading coefficient differs from $A_{\text{ph}}(T)$ by at most 1.6%.

There are T/h steps. We model the measured trace covariance asymptotically as

$$E_{\text{round}}(h) \simeq Bu\sqrt{T/h},$$

where B absorbs the number and sensitivity of rounded updates per step and details of this state metric. Substituting the phase-only bias and covariance models into the bias and variance identity gives

$$E_{\text{rms}}^2(h) \simeq A_{\text{ph}}(T)^2 h^4 + B^2 u^2 \frac{T}{h}.$$

To estimate B , we fit $y_i \simeq B^2 x_i$ by least squares through the origin, where y_i is the unbiased trajectory-covariance trace and $x_i = u_i^2 T/h_i$. The fit uses $s = 10, 12, 20, 24$ at $T = 100$ and $h_i \leq 0.2$. It excludes all $s = 16$ observations.

Differentiating with respect to h and setting the result to zero yields

$$\begin{aligned} 4A_{\text{ph}}(T)^2 h^3 - B^2 u^2 T h^{-2} &= 0, \\ h_\star &= \left(\frac{B^2 u^2 T}{4A_{\text{ph}}(T)^2} \right)^{1/5} \\ &= \left(\frac{144B^2}{T} \right)^{1/5} u^{2/5}. \end{aligned}$$

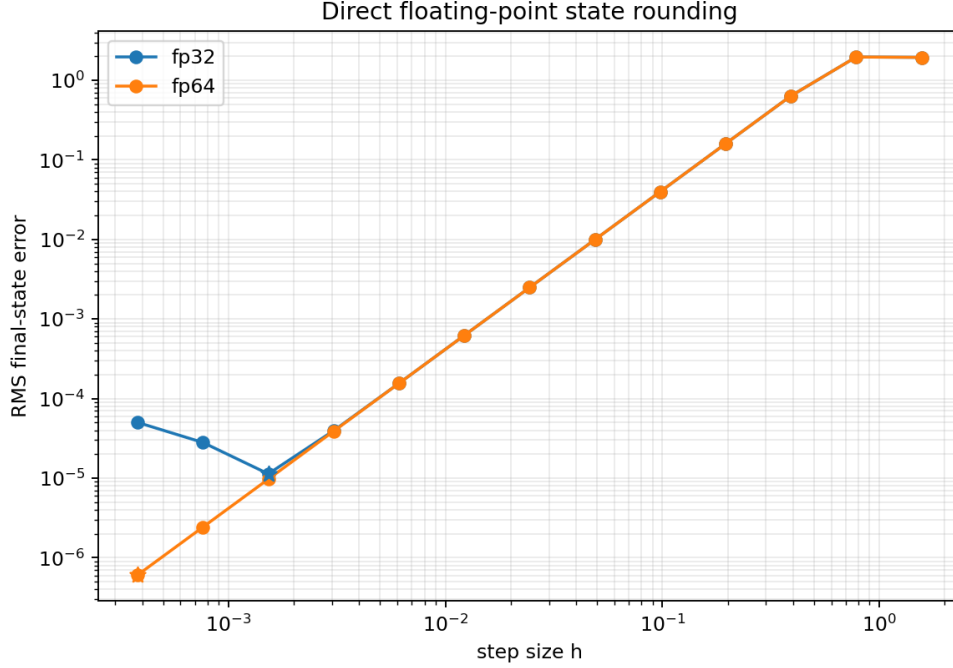


Figure 1: Direct state rounding at $T = 100$, averaged over 128 initial phases on the unit-radius circle. Stars mark grid minima.

The symbol h_* denotes the continuous minimizer of this asymptotic model. We use h_{grid} for a minimum selected from the tested timestep grid. Writing the predicted scaling as $h_* \propto u^{\beta_s} T^{\beta_T}$ gives $\beta_s = 2/5$ and $\beta_T = -1/5$. The precision experiment estimates B . The duration sweep tests transfer of that estimate and the predicted duration exponent.

4 Results

4.1 Direct FP32 crossover

For FP32, the minimum sampled RMS final-state error is 1.13×10^{-5} at $h = 1.5259 \times 10^{-3}$. At the next finer step, $h = 7.6294 \times 10^{-4}$, error rises to 2.80×10^{-5} . at $h = 3.8147 \times 10^{-4}$ it is 5.01×10^{-5} . Two successive refinements therefore worsen this metric after the minimum. FP64 continues to improve through the finest tested step, reaching 6.06×10^{-7} , so its crossover is below or unresolved by this grid (Figure 1).

4.2 Stochastic-rounding crossover

The grid-selected minimizing steps for $s = 10, 12, 16, 20, 24$ are respectively 4.883×10^{-2} , 3.453×10^{-2} , 1.221×10^{-2} , 2.158×10^{-3} , and 1.079×10^{-3} . For each width, let h_{grid} be the step that minimizes estimated risk on the tested grid. A least-squares fit of $\log h_{\text{grid}}$ against $\log u$, excluding $s = 8$, gives $\hat{\beta}_s = 0.419$. We bootstrap 2,000 ensembles, resampling the 128 trajectories independently at every precision and timestep, recomputing covariance risk, and reselecting each minimum. The percentile 95% interval for $\hat{\beta}_s$ is $[0.392, 0.448]$, which includes the predicted exponent $\beta_s = 0.4$. The per-width optimum intervals are grid-selection intervals under this same bootstrap. Fitting the exact matrix bias and measured covariance trace gives $B = 0.6388$ using $s = 10, 12, 20, 24$. Without refitting B , the asymptotic prediction at held-out $s = 16$ is $h_* = 0.01065$, versus the precision-sweep grid minimum $h_{\text{grid}} = 0.01221$ (12.8% relative step error).

The $s = 8$ point is excluded from the primary exponent fit as the coarsest simulated precision. Including it changes the slope only from 0.419 to 0.418. The minima are selected from a finite, half-octave step grid, so the fitted slope is a resolution-limited estimate. Figures 2 and 3 show the error curves and the

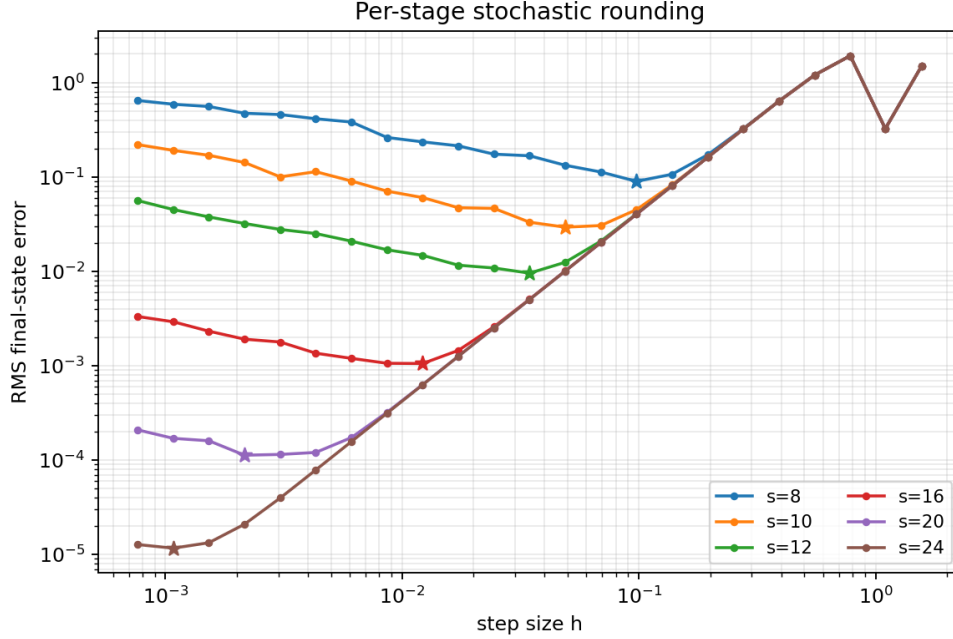


Figure 2: Estimated RMS final-state risk under stochastic rounding at $T = 100$. Each curve uses the exact matrix bias and a sample covariance trace from 128 trajectories per grid point. Stars mark grid minima.

Table 1: Asymptotic fixed- B predictions and independently measured duration-grid minima at $s = 16$. The interval is the percentile trajectory-bootstrap interval for the selected grid minimum. Relative difference is $(h_{\text{grid}} - h_*)/h_*$.

Duration T	Predicted h_*	Grid minimum h_{grid}	Bootstrap 95% interval	Relative difference
25	0.01405	0.01333	[0.01111, 0.01429]	-5.1%
50	0.01223	0.01250	[0.01053, 0.01250]	+2.2%
100	0.01065	0.01000	[0.00833, 0.01176]	-6.1%
200	0.00927	0.00909	[0.00741, 0.01000]	-1.9%
400	0.00807	0.00769	[0.00690, 0.00870]	-4.7%

selected optima.

4.3 Independent duration scaling

At held-out significand width $s = 16$, we test $T \in \{25, 50, 100, 200, 400\}$ with 512 independent rounding trajectories per step on the common widened grid. The bootstrap resamples the independently generated trajectory ensemble separately at every timestep and duration, while preserving paired state components within each trajectory. At each bootstrap draw we re-estimate covariance and reselect each grid minimum. Across 2,000 draws, the fitted exponent in $\log h_{\text{grid}}$ versus $\log T$ is $\hat{\beta}_T = -0.205$, with percentile 95% interval $[-0.254, -0.144]$. This interval contains the predicted $\beta_T = -0.2$. These intervals include trajectory-sampling and grid-selection uncertainty conditional on the tested grids. They do not include uncertainty in the separate fit of B .

The asymptotic fixed- B predictions below use $B = 0.6388$, fitted from the $T = 100$ precision sweep at $s = 10, 12, 20, 24$, and evaluate $h_*(T) = (144B^2u^2/T)^{1/5}$ at $u = 2^{-16}$. Neither the calibration nor these predictions use duration-sweep observations. Observed minima are selected independently from the common grid.

This comparison tests calibration transfer in absolute scale across duration, in addition to agreement of fitted exponents (Table 1 and Figure 4).

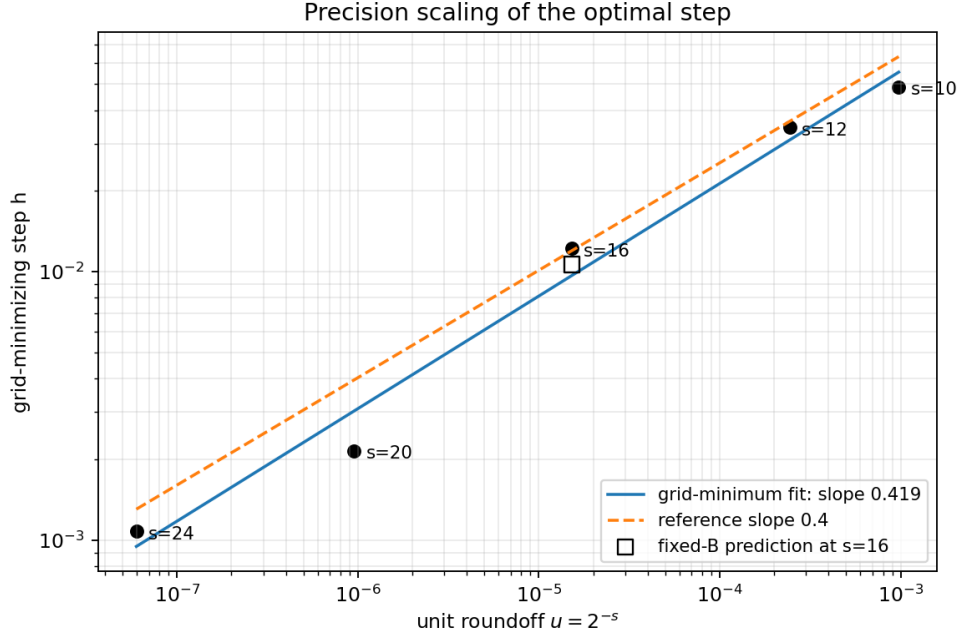


Figure 3: Grid-minimizing steps versus unit-roundoff scale. The fitted slope uses $s = 10, 12, 16, 20, 24$. The dashed reference line shares its intercept with that fit. The open square is the asymptotic fixed- B prediction calibrated without $s = 16$.

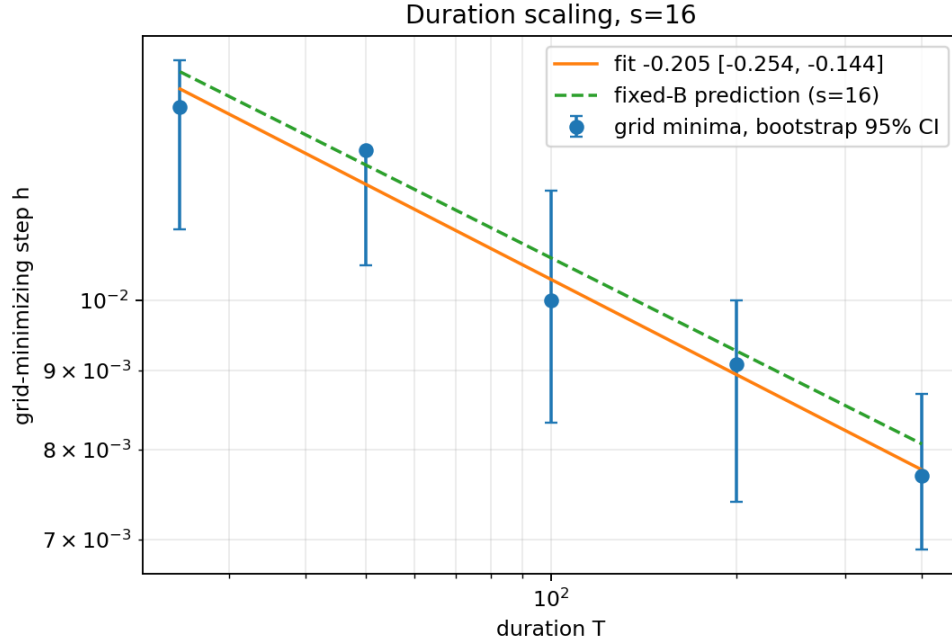


Figure 4: Observed grid minima with 95% trajectory-bootstrap intervals and the asymptotic fixed- B prediction at $s = 16$. The same absolute timestep grid is used at each duration.

5 Discussion and limits

The controlled experiment retains timestep coefficients in FP64 and rounds only the three evolving state outputs per step. Its mean bias is computed exactly from the Verlet matrix, while stochastic variance comes from replicate covariance. The widened duration test uses one common absolute h grid at all five durations. The $s = 16$ observations are excluded from fitting B and held out for this test. Its point estimate and bootstrap interval are compared with the prediction in the results. The phase-only fixed- B curve predicts a continuous optimal step at each duration. All five observed grid minima are within 6.1% of the predictions, and all five predictions lie inside the bootstrap intervals for the observed minima. Separately, the fitted fixed- B optimum is an out-of-sample precision check: it is 12.8% lower in step size than the observed $s = 16$ precision optimum.

The relation to previous roundoff studies is specific. Kehlet and Logg estimate ODE solution error a posteriori by propagating local errors, including roundoff, through an adjoint sensitivity calculation [4]. Here, the deterministic bias is available in closed form, allowing us to isolate stochastic covariance and test an explicit optimal-step prediction. Connolly, Higham, and Mary analyze stochastic rounding probabilistically and emphasize mean independence, which is weaker than mutual independence [3]. Fresh independent draws are used at each rounded state update in our implementation, but recurrence propagation can induce dependence in final-state errors. We measure the full trajectory covariance instead of assuming that cross terms vanish. This is a focused quantitative benchmark that complements these broader analyses.

The direct FP32 experiment and configurable stochastic-rounding experiment answer different questions. The former uses standard round-to-nearest and shows a clear crossover in this fixed oscillator. The latter isolates unbiased random rounding and supports a fitted law. Neither result alone proves that hardware roundoff behaves as independent stochastic rounding, nor that the fitted B transfers to other systems, integrators, durations, or observables. Real arithmetic also has finite exponent ranges and operation-level rounding patterns that this configurable format does not reproduce.

6 Conclusion

For $T = 100$, direct FP32 velocity Verlet has a state-error minimum near $h = 1.5 \times 10^{-3}$ followed by increasing error under further refinement. Under the specified simulated stochastic-rounding operator, the exact bias and variance decomposition and held-out common-grid duration sweep give exponents $\hat{\beta}_s = 0.419$ versus u and $\hat{\beta}_T = -0.205$ versus T . The modified-frequency expansion and covariance-growth model give the long-duration phase approximation

$$h_* = \left(\frac{144B^2u^2}{T} \right)^{1/5} \propto u^{2/5}T^{-1/5}.$$

The fit $B = 0.6388$, the precision exponent interval $[0.392, 0.448]$, its held-out $s = 16$ predictions, and duration-slope interval $[-0.254, -0.144]$ quantify this controlled setup while keeping the limits of the asymptotic model explicit.

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